

Clinical Practice Guidelines for the Use of Transcranial Direct Current Stimulation in Psychiatry

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INTRODUCTION

Transcranial direct current stimulation (tDCS), a safe and non-invasive neuromodulation technique, has re-emerged over recent years with several technical optimizations. Given the limits of extant therapeutic options in psychiatry, mainly because of its tolerability and safety profile, tDCS has elicited significant interest in clinical research studies in psychiatry, neurology, and several other medical specialties. These studies are also matched with cutting-edge investigative neuromodulation research using tDCS that has revealed critical insights advancing our knowledge about the brain in health and disease.^[1] In psychiatry, tDCS has been evaluated in treating major depressive disorder, schizophrenia, alcohol use disorder, obsessive-compulsive disorder, mild cognitive impairment/dementia, and several other disorders. Given its portability and cost-effectiveness, tDCS offers the option of the remotely supervised, home-based (domiciliary) application as well.

tDCS - DEFINITION

tDCS uses the application of low-intensity, direct (time-invariant) current (usually in the range of 1-2 milliamperes [mA]). This non-invasive neuromodulation technique, if administered as per recommended standard operating procedures, is extremely safe. The current delivery is ensured through the placement of electrodes (25-35 cm² size [i.e., 5 X 5 cm or 7 X 5 cm]) that are made of bioconducting material (e.g., conductive rubber) placed on the scalp (corresponding to the underlying target brain area) leading to polarity-specific neuromodulation and adaptive neuroplasticity changes in the neural regions [Figure 1].

tDCS - MECHANISM OF ACTION

The therapeutic utility of tDCS in disorders can be best understood when both the neuroplasticity mechanisms and how tDCS modulates those mechanisms are adequately deciphered. Studies have noted that the observed changes on account of tDCS are secondary to the adaptive neuroplasticity of the human brain.^[2] Simply speaking, the mechanism through which tDCS can modify neuroplasticity

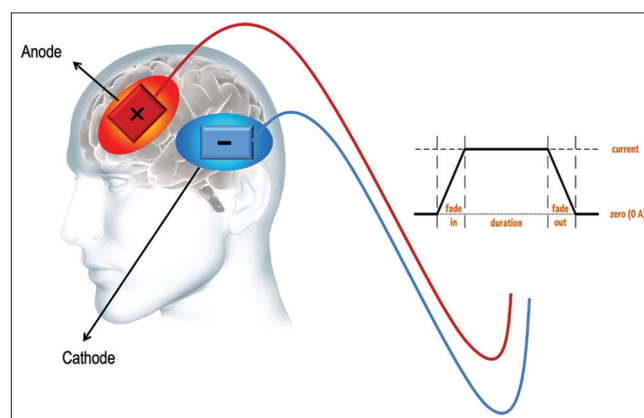


Figure 1: Schematic illustration of tDCS application

is either by increasing or decreasing neuronal conductivity, differentially acting on the neuronal sites, modulating the local blood flow, and brain-derived neurotrophic factor (BDNF)-dependent mechanisms as well as glutamatergic, GABA-ergic/other neurotransmitter-mediated effects.

Alteration of the neuronal resting membrane potential by varying the cation permeability is postulated to be one of the primary mechanisms through which tDCS is claimed to act. When a strong depolarizing signal is applied, it leads to more than the usual influx of calcium ions pre-synaptically. A more significant influx of calcium ions results in a greater release of glutamate post-synaptically, which subsequently causes extensive NMDA receptor activation. This cascade eventually causes an increased calcium influx post-synaptically, activating protein kinases responsible for the phosphorylation of AMPA receptors. Phosphorylation of AMPA receptors further activates more AMPA receptors resulting in further cation permeability in the postsynaptic neuron and better synaptic conductivity. The effects of tDCS on neuroplasticity can be summarized as follows: tDCS causes an increase in synaptic conductivity, both immediate and long-term. The polarity of the tDCS plays a vital role in determining the local effects of the procedure, both at the regional and neuronal levels. tDCS has been shown to induce long-lasting synaptic potentiation via augmented BDNF secretion. The long-term effects are also

believed to be secondary to gene transcription secondary to tDCS. tDCS is also likely to show effects by polarity-based modulation of local blood flow. Polarity-specific effects are time-dependent, with longer duration (generally more than 40 minutes) of stimulation session provoking compensatory mechanisms and reversal of effects.^[3,4]

tDCS: DEVICE ASPECTS

There are several types of tDCS devices. In each one, electrodes are connected to a device capable of delivering a constant low-intensity direct current (0.5 to 3.0 mA). In conventional tDCS, two large conductive siliconized rubber electrodes (typically $7 \times 5 \text{ cm}^2$), anode and cathode, complete the circuit. This administration is polarity-specific in effect, in which inhibitory stimulation at one region is counterbalanced for an excitatory stimulation of equal intensity at another region and vice versa. With this montage arrangement, we can deliver bipolar stimulation; that is, the nature of stimulation is anodal and cathodal in effect because of the electrode type and montage configuration.

The tDCS devices deliver constant current – the intensity of current remains steady over time (e.g., 1 mA or 2 mA). As per Ohm's law, the current intensity is directly proportional to voltage and inversely related to the resistance in the circuit.

Ohm's law: Voltage = Current Intensity x Resistance
[$V = I * R$]

Living biological tissue reacts to electric current as a way of adaptation, along with the flow of tissue fluids and alters the resistance being offered in the circuit. Hence, effective resistance in the circuit involves biological tissue as a combination of ohmic resistance and reactance. This effective resistance is called impedance.

To keep the current constant, where changes in impedance happen dynamically, the device adjusts the voltage at every given time point. As a safety precaution, most medical-grade devices maintain a cutoff of impedance (generally 10–15 kohms) to avoid voltage surge. When the resistance increases beyond a certain threshold or when voltage reaches its limit, the device pauses/terminates the stimulation. Resistance indicators would generally be available and display the contact conditions between electrodes and the scalp during the sessions. Thus, it is recommended to use only those tDCS devices that are certified for human administration that complies with the required safety standards.^[5,6]

tDCS: STANDARD OPERATING PROCEDURES

Checking for contraindication to tDCS during subject recruitment

1. Choose the patients by ascertain the indication and any necessary precautions for tDCS administration [Table 1].

2. Explain to the subject the tDCS procedure in detail. Use of video would assist in annihilating the apprehensions.
3. Ask the subject the following questions in the screening questions [Table 2] to enquire about the potential factors influencing the safe application of tDCS procedure.

Although there are no absolute contraindications for tDCS, administrators had to be cognizant of the above factors while planning the sessions. Past history of adverse effects should enable the administrators to take appropriate steps, as discussed later. Any brain-related injury, surgery, or space-occupying lesions can affect how the effect of electrical distribution and its consequent effects. History of epilepsy can theoretically increase the risk of seizures, specifically if the seizurogenic foci underlie the anode. Though tDCS has been administered safely in individuals with metallic/electrical implants, the distance from the stimulation site and the sensitivity of these devices to the electric field has to be considered. As medications can be a major confounding variable influencing the tDCS effects, it is always prudent to document it. Few patients develop headaches after tDCS, and awareness of the details of headache history in the patients will assist in its appropriate management.

Table 1: tDCS indications and precautions

Indications	Precautions
1. Major depressive disorder	a. Structural head injury
2. Persistent auditory hallucinations in schizophrenia. Possibly for positive and negative symptoms.	b. Epilepsy in patient/family
3. Craving in alcohol dependence and tobacco smoking: Relapse prevention	c. Scalp injury/skin lesions
4. Obsessive-compulsive disorder	d. Implanted medical devices
5. Mild cognitive impairment and dementia	e. Foreign body in head/eyes
	f. Past history of adversities with tDCS/rTMS

Table 2: Screening questions to identify potential factors influencing tDCS procedure

Questions	Remarks
Had any adverse reaction to TMS/tDCS, if received earlier?	
Had a seizure/epilepsy?	
Had an unexplained loss of consciousness?	
Had a stroke?	
Had a serious head injury?	
Had a surgery to your head?	
Had any brain-related, neurological illnesses?	
Had any illness that may have caused brain injury	
Do you suffer from frequent or severe headaches?	
Do you have any metal in your head (outside the mouth) such as shrapnel, surgical clips, or fragments from welding?	
Do you have any implanted medical devices such as cardiac pacemakers or medical pumps?	
Are you taking any medications?	
Are you pregnant?	
Does anyone in your family have epilepsy?	
Do you need any further explanations on tDCS/HD-tDCS or its associated risk?	

Preparation^[4]

Written informed consent has to be taken.

- Information regarding presence of minimal evidence for acute short-term efficacy and absence of strong evidence for long-term clinical efficacy of tDCS in the above-mentioned indications has to be clearly stated.
- Safety of multisession tDCS in clinical patients can be reassured. The provision of aborting the session in the event of intolerable side effects would further annihilate the patient's safety concerns.

Instruction to visit for tDCS with the dry, clean, non-oily scalp for tDCS session should be provided.

Patients (and caregivers) have to be informed that fasting or other lifestyle changes are not needed for tDCS administration.

Materials and their description (see Table 3 for reference)

1. Check if you have all materials needed before starting the procedure.
2. The tDCS device is a battery-driven current generator capable of providing constant current stimulation to the brain with a maximum output (of ± 2.5) in milliamperes (mA) range. It operates on a rechargeable power bank.
3. Electrodes used for tDCS are conductive rubber electrodes. These can deliver DC of either polarity - anode and cathode depending on how they are plugged into the machine.
4. Measurement tape and skin marker can be used to mark the desired location on the subject's scalp. The measuring tape can be further used to ascertain adequate distance between the anode and return cathodal electrode as per the study protocol requirements (minimum 3-finger distance).

Table 3: List of materials for conventional tDCS

Materials**tDCS Device and related components**

- Battery-operated tDCS device
- Rubber electrodes (minimum 2)
- Specially designed sponges meant to keep the electrodes in the scalp
- Rubber bands (non-conductive)
- Cable to connect the rechargeable battery (power bank) with the tDCS machine
- One rechargeable battery
- An adapter for recharging the power bank battery

Other Materials

- Saline (9 mmol)
- Conductive electrode paste (if needed to decrease impedance by effectively parting the hair)
- Suction pipette and trough to keep the saline.
- Cotton to clean the contact surface
- Measurement tape
- Marker (for marking measurement marks on subject's scalp)
- Comfortable chair (without exposed electrically conductive surface)
- Tissue for wiping the skin region where electrodes were placed after the session

5. A comfortable chair is required to seat the subject in relaxed manner throughout the preparation and administration of tDCS procedure.
6. Tissue or paper towels can come in handy for cleaning off either electrode due to excessive saline or cleaning the subject's scalp after administration.

Pre-administration preparations

1. Turn on the tDCS device. Ensure the device has enough power for completion of the session.
2. Visually inspect the rubber electrodes for signs of wear and tear.
3. Place the rubber electrodes in sponge casings to improve tolerability and reduce adverse events like tissue injury. Never place the electrodes directly on the scalp.
4. Apply the non-conductive water-proof bands for holding electrodes securely on the subject's head.
5. The placement of electrodes is described in the following sections.

Administration procedure

1. Seat the subject comfortably in a chair.
2. Thoroughly inspect the subject's scalp for signs of skin lesions, cuts, signs of inflammation or other cutaneous abnormalities.
3. Localize the stimulation target regions on the subject's head
 1. 10-20 EEG system
 2. Using tools like BeamF3
 3. Neuro-targeting using structural with/without functional magnetic resonance imaging
4. Mark the point on the subject's scalp that corresponds to target locations.
5. Part the hair at this marking. EEG paste can be used to keep hair parted - thick hair can cause higher impedance.
6. Switch on the device before placing the electrodes on the scalp surface. This is to avoid sudden surge of current in the circuits that can lead to adverse effects.
7. During the electrode placement, make sure that the smooth surface of the electrode (and not the wired-connected electrode surface) is in contact with the scalp.
8. Sponge preparation: Add saline to the sponge (around 6 ml on either sides) to make it damp. Ensure it is sufficiently damp and not dry to be properly conductible. It should also not be dripping with saline, which may result in the shortening of circuit. Once prepared, place the electrode inside the sponge. Inspect the sponges for reusability in multisession administrations.
9. Carefully place the cathodal and anodal electrodes kept inside the sponge case on the mark for desired/marked target regions on the cleaned subject's scalp at an appropriate orientation. For example, tDCS electrodes

for auditory hallucinations should be placed in a horizontal orientation with 7 cm as the length for the left temporoparietal junction and a vertical orientation with 5 cm as the length for the left dorsolateral prefrontal cortex. The wire connected to the electrode should be posteriorly directed in the attachment.

10. Ascertain the distance between the two electrodes is minimum 7 cm (3 finger distance).
11. Check for the subject's comfort level with the attached headbands over the electrodes (This can be ascertained by asking the patients, "Is the setup too tight?")
12. Set-up the electrical parameters, including peak intensity, duration of stimulation, ramp-up and ramp-down rate/duration.
13. Initiate the treatment. Ensure the impedance is below 10–15 k Ω - with further increase in resistance, the machine will auto-terminate the session. *
14. Check for any sensations and reassure the pain will reduce in a few seconds with the completion of ramping up and development of tolerance.

*Note: In situations of high impedance or in case of more pain, check for the following causes:

- a. Check if the electrodes are in full contact with the scalp. Make appropriate changes to establish better contact.
 - b. Check if the saline is too less. Add saline as required.
 - c. Check if the saline is too much and the current is being shunted. Use tissue to remove the extra dripping saline.
 - d. Check if the hair parting. Remove the electrodes entirely, part the hair, apply EEG paste, and then re-attach the electrodes. In most circumstances, the above steps will resolve the impedance issues.
15. After the session is over, remove the electrodes. Switch off the machine only after the electrodes are removed.
 16. Electrodes need to be removed from the sponge pads. Wipe the electrodes with the tissue (since dry saline over the electrodes can damage them and decrease its shelf life.)
 17. Clean the sponges with running water and allow them to completely dry before the next session. Ensure the salt deposits won't stay for the next session.

After the tDCS procedure

1. Carefully inspect the skin regions where electrodes were placed for signs of skin irritation and/or skin damage. Any skin deterioration should be addressed in a medically appropriate way.
2. Document the session-specific details in the session record sheet.
3. Enquire from the subject about any possible side effects and fill in the details in the side-effect record sheet after every session.
4. Inform the subject and their relative (or caregiver) about the timings of the next session and brief them if there were any major issues during or from the session (like

appearance of skin lesion after the session, repeated sudden cessations throughout the session, difficulty in initiating stimulation due to abnormally high impedance, persistent moderate to severe side effect, etc.)

5. Check the power status. If required, charge the device.

Precautions

1. Make sure that the subject takes regular head-bath, refrains from oiling hair/scalp, and has a clean scalp when he/she comes for tDCS session.
2. Carefully inspect skin regions where electrodes are placed before and after every session.
3. The subject should be relaxed, comfortable, and awake throughout the tDCS procedure.
4. Uncontrolled interference with ongoing cortical activity during tDCS sessions should be avoided. Intensive cognitive effort or doing unnecessary motor activities should be avoided during the session.
5. Do not turn on the tDCS device before setting up or the electrodes for safety reasons. In a similar vein, the device should be switched off after the administrators have unmounted the electrode set-up, following completion of the stimulation.
6. Before starting the stimulation make sure that the electrodes wires are uncoiled or untangled where they connect to the machine. Coiled wires may interfere with and increase the overall resistance in the circuit.

Monitoring of adverse effects due to tDCS

tDCS is a safe, well-tolerated intervention if applied using the standard procedures and protocols. The safety report is based on the stimulation parameters that are commonly examined in patients. The adverse events are assessed using a checklist/questionnaire which evaluates the severity as well as the grade of attribution of these adverse effects to the tDCS^[7,8] [Table 4].

APPLICATION OF tDCS IN PSYCHIATRY DISORDERS

Schizophrenia

Schizophrenia is a debilitating, chronic neuropsychiatric disorder, which also is a leading cause of disability burden. The symptom components of schizophrenia involve delusions, hallucinations, significant cognitive and motivational impairments. Despite treatment with best of the available antipsychotic medications, about 30% of schizophrenia patients show partial or no clinical improvement and they persist to have symptoms. Significantly, treatment resistance contributes to about 80% of total healthcare cost burden due to schizophrenia. Contextually, alternative paradigms that involve non-invasive brain stimulation techniques attract increasing application in treating resistant symptoms in schizophrenia patients. Among several neuromodulatory

Table 4: tDCS adverse effect questionnaire

Adverse effect	Severity	Related to tDCS
Headache		
Neck pain		
Scalp pain		
Tingling		
Itching		
Burning sensation		
Skin redness		
Sleepiness		
Trouble concentrating		
Acute mood changes		
Skin lesion		
Disturbed visual perception		
Discomfort (during tDCS)		
Dizziness		
Pressure		
Flashes (Phosphenes): during initiation	(Yes=1/No=0)	
Flashes (Phosphenes): during termination	(Yes=1/No=0)	
Other (Please specify)		
Severity: 1-Absent, 2-Mild, 3-Moderate, 4-Severe. Related to tDCS: 1-No, 2-Remote, 3-Possible, 4-Probable, 5-Definite		

techniques, tDCS has been gaining an increasing evidence base to support its clinical utility in treatment-resistant schizophrenia.

The tDCS protocols in schizophrenia have been informed by neuroimaging studies that demonstrated association between left temporoparietal region hyperactivity and auditory hallucinations as well as relationship between hypofrontality and the pathogenesis of negative symptoms. The commonly used tDCS electrode montage applies cathodal stimulation over the left temporoparietal junction and anodal stimulation over the left prefrontal cortex to target auditory hallucinations and negative symptoms, respectively.

A pioneering study that applied a randomized, double-blind sham-controlled design on tDCS for treatment-resistant auditory hallucinations in schizophrenia demonstrated a 30% improvement in hallucination scores which persisted for about 3 months. Comparable effects were reported in a couple of open-labeled studies and multiple case reports. Beyond the ameliorative effects on auditory verbal hallucinations, clinical research studies have described evidence to support beneficial effects of tDCS on negative symptoms, other general symptoms related to psychopathology as well as illness awareness.^[9-12]

Major depressive disorder

Major depressive disorder (MDD), the leading cause of disability burden, is a healthcare challenge. The myriad dimensions of symptoms, coexistent psychiatric and other medical morbidities further add to the complexity toward management. The recalcitrant symptoms in MDD warrant newer treatment modalities, and tDCS is increasingly seen as a useful avenue.

A key dysfunctional brain region in MDD is the prefrontal cortex – especially the dorsolateral region – of the left side (DLPFC); hypoactive left DLPFC and hyperfunctional right DLPFC is postulated as one of the contributory components to MDD. This offers a suitable opportunity to apply anodal current to enhance the hypoactive left DLPFC and cathodal current to optimize the right DLPFC.

MDD is perhaps the most extensively examined psychiatric disorder with treatment studies using tDCS. Summary evidence from these large numbers of studies suggest better efficacy of tDCS for first-episode/early course MDD and lesser benefits in treatment-resistant depression. Increasingly, evidence supports sustained effects of tDCS in MDD beyond the intervention period. Interestingly, studies have supported the feasibility of home-based application as well. Other special situations that warrant the consideration of tDCS in MDD include pregnancy (especially the first trimester), multiple medical co-morbidities with serious risk even in the context of minor side-effects/drug interactions that render stiff challenges to psychopharmacological approaches.^[9,13,14]

Obsessive-compulsive disorder

Obsessive-compulsive disorder (OCD), with a lifetime prevalence of 1–3%, is among the leading causes of neuropsychiatric disability. Animal models and human neuroimaging studies implicate a dysfunction in the parallel and partially segregated cortico-striato-thalamo-cortical (CSTC) circuits as well as the fronto-limbic circuits in the pathogenesis of OCD. These circuits modulate various cognitive, affective, and motivational processes, which are affected in patients with OCD. Recent studies also suggest cerebellar involvement in pathogenesis. Although selective serotonin reuptake inhibitors (SSRI) and cognitive-behavior therapy (CBT) are the first-line treatments for OCD, a sizable proportion of patients do not respond adequately to these interventions. Invasive and non-invasive neuromodulatory interventions have been attempted to modulate the above circuits in patients with treatment resistant illness, usually as an augmentation strategy.

Functional imaging studies have shown hyper as well as hypoactivity in various cortical regions and cerebellum, which are potential targets of non-invasive neuromodulatory interventions such as tDCS. However, it is unclear whether the dysfunction is primarily related to the pathogenesis or is compensatory. Hence, both anodal and cathodal tDCS have been attempted, with mixed results. Among the various protocols, RCTs have shown significant improvement with active tDCS protocols targeting the supplementary motor area/pre-supplementary motor area (SMA/pre-SMA) and the right cerebellum as compared to sham stimulation. Although the SMA/pre-SMA target has been studied by independent groups showing positive results, studies have employed varied targeting strategies, stimulating either on the left

side or bilaterally. Anodal as well as cathodal tDCS over the SMA have shown encouraging results in sham-controlled studies. A crossover study showed superior response to cathodal compared to anodal stimulation. A recent evidence-based guideline recommended anodal tDCS with Level-C recommendation. Thus, there is a need for larger studies to compare the efficacy of cathodal vs anodal tDCS targeting SMA/pre-SMA. Anodal tDCS over cerebellum with cathodal stimulation over left orbitofrontal cortex has also been shown to be helpful in two independent RCTs. Sham-controlled trials employing anodal tDCS over left DLPFC have shown inconsistent results, although the studies employed varied methodologies. Protocols targeting the orbitofrontal cortex have not been evaluated in sham-controlled studies. There is also preliminary evidence for augmentation of CBT with anodal tDCS over medial prefrontal cortex (mPFC). Overall, systematic evidence exists for protocols employing anodal or cathodal stimulation of SMA/pre-SMA and anodal stimulation of the right cerebellum. Larger studies are required to confirm these findings as well as to evaluate the diverse targeting strategies and stimulation protocols.^[9,15,16]

Tourette syndrome

Tourette syndrome is a childhood-onset neuropsychiatric condition characterized by motor and vocal tics. The disorder wanes off during adolescence in most individuals, although dysfunctional tics persist in a subset of patients. Cathodal tDCS targeting the supplementary motor area has been employed as an augmentation strategy in pharmacotherapy resistant Tourette syndrome. However, the available evidence is preliminary in the form of case reports/series or single-session treatment. Given the preliminary evidence for low frequency rTMS over the same region in Tourette syndrome, cathodal tDCS may be a promising protocol that requires evaluation in larger systematic studies. At present, the evidence is preliminary at best.^[9]

Anxiety disorders

Anxiety disorders are among the most common and disabling psychiatric disorders. CBT and SSRIs are the first-line treatments. Mesocortico-limbic pathways involving the DLPFC, anterior cingulate cortex (ACC), amygdala, and hippocampus are implicated in the pathogenesis. Emotional regulation, fear processing, and extinction are neuropsychological functions that may be deranged in these disorders. Non-invasive neurostimulation of cortical regions modulating these functions including dorsolateral prefrontal cortex (emotional regulation) and ventromedial prefrontal cortex (fear extinction) may thus be helpful. tDCS over these regions has been attempted as a standalone treatment or as an augmenter for CBT/other psychological interventions. Similar to depression, anodal stimulation of left DLPFC and/or cathodal stimulation of right DLPFC are the commonly employed protocols, while mPFC anodal stimulation is sometimes used for augmenting

fear extinction in exposure therapies. Preliminary evidence from single studies on individual disorders have shown some encouraging results, although some studies have not shown benefit. In the absence of additional data, tDCS may currently be recommended only as experimental treatments for these conditions.^[9]

Substance use disorders

Substance use disorders are a group of highly prevalent chronic relapsing conditions, characterized by craving or irresistible urges to take particular substances, emergence of physical as well as negative emotional affective state in the absence of the substance and inability to cut down on the use. The above manifestation is modulated by distinct neuronal pathways - reward/incentive salience (basal ganglia), negative emotional (extended amygdala/habenula), and craving/executive function pathway (prefrontal cortex/insular), respectively. Although standard pharmacological and psychological interventions are effective, they have their limitations in terms of acute and long-term efficacy. tDCS has been employed to boost the outcomes, sometimes combined with other psychological interventions. The executive function pathway involving prefrontal cortex is particularly amenable to non-invasive stimulation and has been the target in many tDCS studies.

For alcohol use disorders, a protocol involving anodal tDCS of right DLPFC and cathodal tDCS of left DLPFC has shown the most consistent results in sham-controlled trials. This protocol has been found to reduce craving, long-term relapse and improve network efficiency/inter-regional connectivity in the brain. The abstinence efficacy of the above protocol can be augmented employing alcohol-specific inhibitory control training. It is to be noted that the above protocol is reverse of that used in depression, where anodal tDCS of left DLPFC is combined with cathodal stimulation of right DLPFC. The “depression protocol” has not been found to be helpful in sham-controlled clinical trials in alcohol use disorders when combined with bias modification. Anodal tDCS of the right inferior frontal gyrus has not been found to be helpful in mindfulness-based relapse prevention. Thus, the most promising tDCS protocol for alcohol use disorders involves anodal stimulation of right DLPFC and cathodal stimulation of left DLPFC.

For tobacco smoking, recent network meta-analyses have shown significant effect of bifrontal tDCS as compared to sham stimulation. Meta-analyses and individual RCTs have shown more consistent efficacy on craving and smoking with right DLPFC anodal tDCS and left DLPFC cathodal tDCS. Studies employing the reverse protocol, i.e., left DLPFC anodal and right DLPFC cathodal stimulation, have yielded inconsistent results. There is also preliminary evidence that bilateral cathodal stimulation of fronto-parietal cortices might decrease smoking consumption. Overall, the best available evidence is for anodal stimulation of right

DLPFC and cathodal stimulation of left DLPFC for smoking cessation, which is similar to the protocol for alcohol dependence. A similar protocol has been used for cocaine and methamphetamine use, which have shown inconsistent results. Larger studies are warranted to evaluate the efficacy of tDCS in these conditions.^[9]

Dementia/Mild cognitive impairment

Cognitive disorders in the elderly are yet another condition in which tDCS studies have been conducted. These studies have shown mixed results. However, there is emerging evidence for anodal stimulation of left DLPFC to be beneficial. Moreover, given the safety and tolerability of tDCS as well as other challenges in the elderly population in the context of medical co-morbidities, potential for poor tolerability of psychotropics, tDCS is an attractive option.^[9,17]

Child and adolescent conditions

tDCS is evaluated in ADHD and autism as well as in learning disorders. But the site of stimulation and protocol is varied, and hence, the evidence base is minimal. Larger studies are warranted to evaluate the efficacy of tDCS in these conditions. Also, the long-term consequences of modulation of the evolving brain are not known. Hence, application of tDCS in this population is experimental and with a word of caution.

EVIDENCE-BASED RECOMMENDATIONS

The available evidence for tDCS in most disorders (in Table 5) are as add-on treatment to ongoing pharmacotherapy or psychotherapy.

Multi-session therapy is needed for longer-lasting clinical effects.

tDCS may have role in certain situations like:

- Patients' preference for non-pharmacological agents
- Non-feasibility of first/second line treatments such as geriatric depression with high-risk for medications-related side-effects and where psychotherapy is unavailable.
- Augmentation for faster response

Theoretically, there is an absence of risk of tDCS during pregnancy but needs to be ascertained in clinical trials.

tDCS is mostly evaluated in treatment-resistant or treatment-persistent symptoms and not in treatment-naïve patients.

Evidence related to the safety and utility of domiciliary tDCS and continuation/maintenance tDCS are minimal (non-replicated RCT)

tDCS is being evaluated to enhance cognition across disorders with some evidence.

Addition of cognitive activity/training to tDCS has not shown additive clinical benefits and may be adversarial.

The therapeutic role of advanced transcranial stimulation like high-definition tDCS, neurotargeted stimulation and other forms of electrical stimulations like transcranial alternating current stimulation, transcranial oscillatory current stimulation, transcranial pulsed current stimulation, transcutaneous cranial nerve stimulations is yet unclear due to absence of evidence (no RCTs) or presence of very low levels of evidence (non-replicated RCT of smaller sample sizes).

Psychiatric conditions with absent/inconclusive evidence from RCTs

1. Neurodevelopmental disorders like ADHD and autism
2. Generalized anxiety disorder
3. Tourette's syndrome
4. Substance use disorders: acute intoxication, increase motivation, relapse prevention (except for alcohol and smoking)

SUMMARY AND CONCLUSIONS

Transcranial direct current stimulation (tDCS), a safe and non-invasive neuromodulation technique, has re-emerged over the recent years with several technical optimizations. The applications in psychiatric disorders are on increase. Contextually, this clinical practice guidelines on tDCS in psychiatry summarize the fundamental concepts related to tDCS, standard operating procedures for clinical practice. In addition, a brief overview of the studies reporting effects of tDCS in various psychiatric disorders is presented. Some of the potential options for the therapeutic application of tDCS include major depressive disorder, schizophrenia (especially

Table 5: tDCS protocols for psychiatric disorders with promising evidence from RCTs

Diagnosis	Anode	Cathode	Duration	Sessions
Schizophrenia	Left DLPFC	Left TPJ	20 min	2 per day × 5 days
OCD*	Pre SMA	Right supraorbital	20 min	2 per day × 5 days
Craving (substance-use disorder)	Right DLPFC	Left DLPFC	20 min	1 per day × 5 days
Depression	Left DLPFC	Right DLPFC	30 min	1 per day × 10 days [^]
Dementia/MCI [§]	Left DLPFC	Right supraorbital	20 min	1 per day × 5 days

*In OCD three types of montages: SMA/Pre-SMA anode, SMA/Pre-SMA cathode, and right cerebellar anode are found to be effective. [^]20-30 days of stimulation are attempted in a few large RCT. [§]In dementia, one RCT has used 10 days daily sessions every month for 8 months. OCD: Obsessive compulsive disorder; SUD: Substance use disorder; DLPFC: Dorsolateral prefrontal cortex; SMA: Supplementary motor area; TPJ: Temporoparietal junction; MCI: Mild cognitive impairment

auditory verbal hallucinations), craving in substance use disorders, obsessive-compulsive disorder, and mild cognitive impairment. While tDCS is in its nascent stage with requirement for further research to ascertain rigorous evidence, some of the advantages of this technique – safety, tolerability, ease of administration, portability, scalability, cost-effectiveness as well as potential for home-based applications - makes this neuromodulation technique a promising therapeutic option in psychiatry.

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Conflicts of interest

There are no conflicts of interest.

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